



FINAL VERSION

VERSION FINALE

**Energy management system application program interface (EMS-API) –
Part 453: Diagram layout profile**

**Interface de programmation d'application pour système de gestion d'énergie
(EMS-API) –
Partie 453: Profil de disposition de diagramme**

CONTENTS

FOREWORD.....	4
INTRODUCTION.....	6
1 Scope.....	7
2 Normative references	7
3 Terms and definitions	7
4 Use Cases.....	8
4.1 General use cases for diagram exchange	8
4.2 Simple bay diagram example	10
5 Diagram layout exchange profile.....	12
5.1 General.....	12
5.2 Diagram layout profile classes	13
5.2.1 General	13
5.2.2 Diagram.....	14
5.2.3 DiagramObject.....	14
5.2.4 DiagramObjectGluePoint	15
5.2.5 DiagramObjectPoint.....	16
5.2.6 DiagramObjectStyle	16
5.2.7 TextDiagramObject.....	16
5.2.8 VisibilityLayer	17
5.2.9 Abstract classes – IdentifiedObject	18
5.2.10 Enumerations – OrientationKind	18
6 Graphical rendering	19
6.1 General.....	19
6.2 Single point objects.....	19
6.3 Multiple point objects	20
6.4 Gluing points.....	21
6.5 Diagram object style	22
6.6 Diagram layout exchange process	22
7 Examples	24
7.1 Data instantiation and encoding	24
7.2 Simple bay example use case	26
Annex A (informative) Benefits and format conversion from IEC 61970-453 Edition 1 to Edition 2	28
Bibliography.....	29
Figure 1 – System overview	9
Figure 2 – Bay diagram as rendered	10
Figure 3 – Bay diagram drawn in GIS style	11
Figure 4 – Bay diagram drawn in SCADA style.....	12
Figure 5 – Diagram layout information model	13
Figure 6 – Conventions used for representing object instances and associations	19
Figure 7 – Single point diagram objects	20
Figure 8 – Multiple point diagram objects	21

Figure 9 – Disconnector with glue point	21
Figure 10 – Disconnector and breaker with glue points	22
Figure 11 – Profiles within IEC standards	23
Figure 12 – Disconnector diagram object instantiation	24
Figure 13 – IEC 61970-552 Encoding for disconnector diagram data	25
Figure 14 – Bay diagram example with objects outlined GIS style	26
Figure 15 – Bay diagram example with objects outlined SCADA/EMS style	27

INTERNATIONAL ELECTROTECHNICAL COMMISSION

**ENERGY MANAGEMENT SYSTEM APPLICATION
PROGRAM INTERFACE (EMS-API) –****Part 453: Diagram layout profile****FOREWORD**

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC itself does not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC is not responsible for any services carried out by independent certification bodies.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

This consolidated version of the official IEC Standard and its amendment has been prepared for user convenience.

IEC 61970-453 edition 2.1 contains the second edition (2014-02) [documents 57/1409/FDIS and 57/1430/RVD] and its amendment 1 (2018-11) [documents 57/2038/FDIS and 57/2054/RVD].

This Final version does not show where the technical content is modified by amendment 1. A separate Redline version with all changes highlighted is available in this publication.

61970-453:2014+AMD1:2018 CSV
© IEC:2018

– 5 –

International Standard IEC 61970-453 has been prepared by IEC technical committee 57: Power systems management and associated information exchange.

This edition includes the following significant technical changes with respect to the previous edition:

- a) The SVG elements and its data model have been replaced by the Diagram Layout Package, which is now an integral part of the IEC 61970-301 (CIM) model.
- b) The exchange is in accordance with and is a part of the IEC 61970 profile concept.
- c) A glue point object has been introduced to model explicit connections between graphics elements.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 61970 series, published under the general title *Energy management system application program interface (EMS-API)*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

IMPORTANT – The 'colour inside' logo on the cover page of this publication indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

INTRODUCTION

This standard is part of the IEC 61970 series that define an application program interface (API¹) for an Energy Management System (EMS²).

The IEC 61970-3x series specify a Common Information Model (CIM³): a logical view of the physical aspects of EMS information. The IEC 61970-3x series includes IEC 61970-301, *Common Information Model (CIM) Base*.

This standard is one of the IEC 61970-4x series that define utility control centre component interface specifications (CIS⁴). IEC 61970-4x specifies the functional requirements for interfaces that a component (or application) shall implement to exchange information with other components (or applications) and/or to access publicly available data in a standard way. The component interfaces describe the specific message contents and services that can be used by applications for this purpose. The implementation of these messages in a particular technology is described in the IEC 61970-5x series.

Energy Management Systems employ a variety of schematic and quasi-geographic presentations in their user interfaces. These are sometimes generated automatically, but more often are hand-drawn and require considerable labour to create and maintain. Most of this labour goes into the arrangement, or 'layout' of the power system elements within the overall diagram. When network models are exchanged, as defined in IEC 61970-452 and IEC 61968-13 standards, it is desirable to be able to exchange these layouts.

IEC 61970-453 specifies guidelines for the exchange of diagram layout information for schematic data that is encoded using IEC 61970-552.

¹ Footnote 1 applies to the French version only.

² Footnote 2 applies to the French version only.

³ Footnote 3 applies to the French version only.

⁴ Footnote 4 applies to the French version only.

ENERGY MANAGEMENT SYSTEM APPLICATION PROGRAM INTERFACE (EMS-API) –

Part 453: Diagram layout profile

1 Scope

This part of IEC 61970 is a member of the IEC 61970-450 to 499 series that, taken as a whole, defines, at an abstract level, the content and exchange mechanisms used for data transmitted between control centre components.

Included in this part of IEC 61970 are the general use cases for exchange of diagram layout data, and guidelines for linking the layout definitions with CIM data. Guidelines for management of schematic definitions through multiple revisions are also included.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050, *International electrotechnical vocabulary*

IEC 61970-301, *Energy management system application program interface (EMS-API) – Part 301: Common information model (CIM) base*

IEC 61970-501, *Energy management system application program interface (EMS-API) – Part 501: Common Information Model Resource Description Framework (CIM RDF) schema*

IEC/TR 62541-1, *OPC Unified Architecture – Part 1: Overview and concepts*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60050, as well as the following, apply.

3.1

domain object⁵

instance of a class that models a Real-World Object⁶ with a unique identity

Note 1 to entry: A domain object inherits from a CIM *IdentifiedObject*. A domain object is normally not a diagram object.

3.2

diagram⁷

electronic equivalent of a seamless paper plan

⁵ Footnote 5 applies only to the French version.

⁶ Footnote 6 applies only to the French version.

⁷ Footnote 7 applies to the French version only.

Note 1 to entry: The diagram is an identified container for the diagram objects. Examples of diagrams include substation schematics, transportation or distribution network orthogonal schematics, or pseudo-geographical schematics. A diagram has a well-defined coordinate space.

3.3

diagram object⁸

representation of domain objects or static background

Note 1 to entry: The diagram is composed of diagram objects.

Note 2 to entry: An example for domain objects includes breakers. An example for static background object includes lakes.

3.4

diagram object style

definition of how to render diagram objects possibly based on the state of domain objects

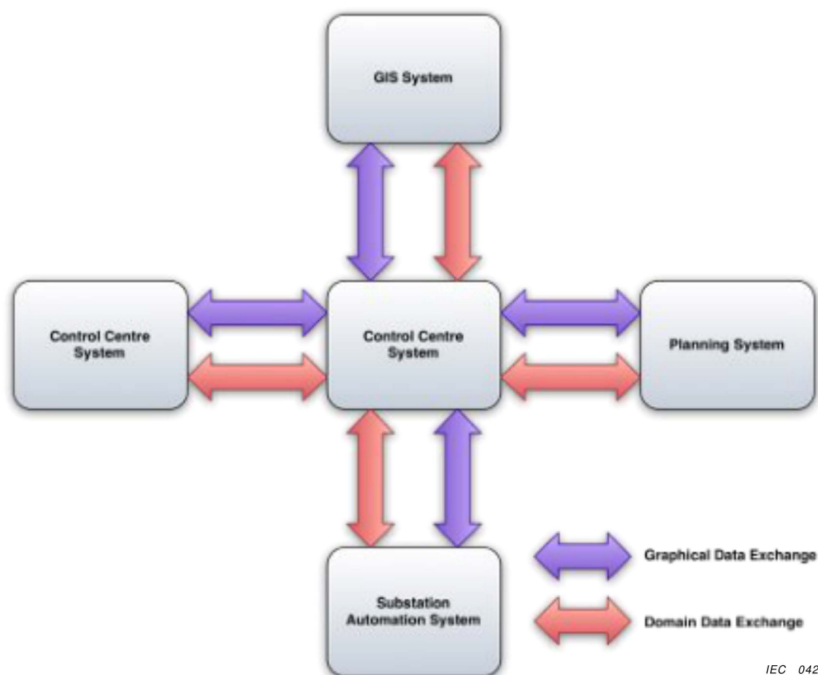
Note 1 to entry: Typically, the diagram object style is resolved in a very specific way for each system.

4 Use Cases

4.1 General use cases for diagram exchange

Figure 1 shows a high-level view of using diagram layout data exchange with potential systems that can make use of the diagram layout data.

⁸ Footnote 8 applies to the French version only.



IEC 0423/14

Figure 1 – System overview

An examination of the use cases for such exchanges revealed that the diagrams being exchanged are not fixed graphic presentations. Instead the diagrams vary considerably in appearance as attributes of CIM objects change, and they support important user interaction. For example, a SCADA one-line in its creator's system is supporting live control of the power system, while in a receiver's system it is merely used for reference so that a neighbour can understand what the system looks like beyond its own border. These variations are typically not easy or appropriate to map between the source and receiving systems because of the degree of difference in the way those systems are designed.

In the specification of diagrams, one common element is that the style of displaying an object of a certain kind is usually defined once and then re-used, but the placement of objects must be adjusted whenever new elements are added to the data model. This placement and maintenance of placement is where most of the labour investment takes place that users would like to preserve. As a result, this standard is limited to the exchange of diagram layouts (meaning the arrangement of CIM objects in a display space) rather than a complete exchange of all characteristics of a graphic presentation.

With this proposed standard, instead of maintaining duplicate schematics for different applications, the schematics are exported by one system and imported by the other system. Diagram layout profile is an extension to the CIM power system model exchange IEC 61970-452, and will be orchestrated along with the existing CIM XML model exchange and updates provided using the existing CIM XML Incremental file format as defined in IEC 61970-552. IEC 61970-552 also describes how payload headers provide information as to how payloads fit together.

This process can be applied for initial schematics construction as well as for continuous maintenance.

The importing system can create its graphics displays from the imported data, or the diagram layout data can serve as additional documentation and means of understanding for the domain data exchange.

4.2 Simple bay diagram example

Diagrams are constructed using different approaches. This subclause illustrates this using a simple bay drawing. Figure 2 shows a typical representation of a bay.

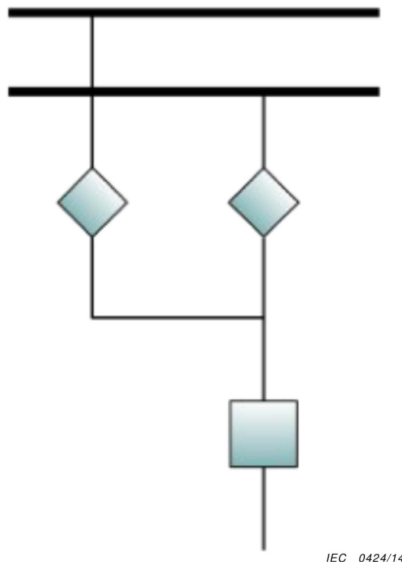
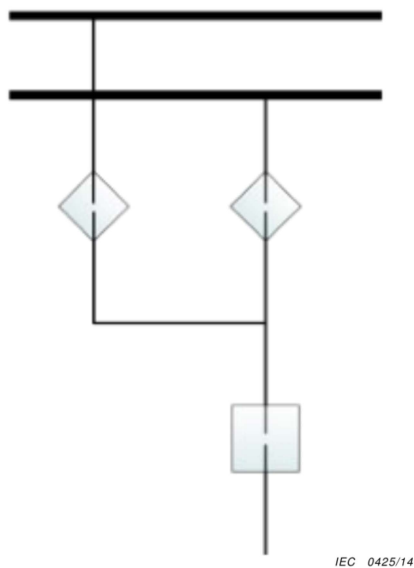


Figure 2 – Bay diagram as rendered

In a GIS system, lines are typically drawn up to the centre point of the symbol. This is shown in Figure 3 (transparent symbol is used for illustration). This construction method is independent of the symbol size, i.e. the drawing looks “pretty” even when the importing system used a smaller symbol size than the original one.

**Figure 3 – Bay diagram drawn in GIS style**

In a SCADA system, lines are typically drawn to align with the terminals of the object represented by the symbol. This is shown in Figure 4 (transparent symbol is used for illustration). This construction method gives a closer representation of the electrical topology, but depends on a certain symbol size.

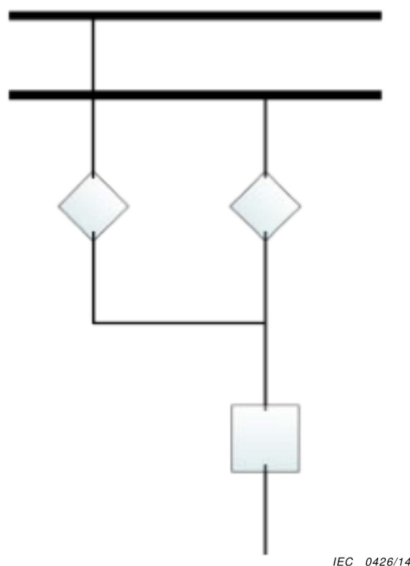


Figure 4 – Bay diagram drawn in SCADA style

5 Diagram layout exchange profile

5.1 General

Profile namespace: <http://iec.ch/TC57/61970-453/DiagramLayout/3#>

This standard specifies an exchange profile for schematics with the following characteristics.

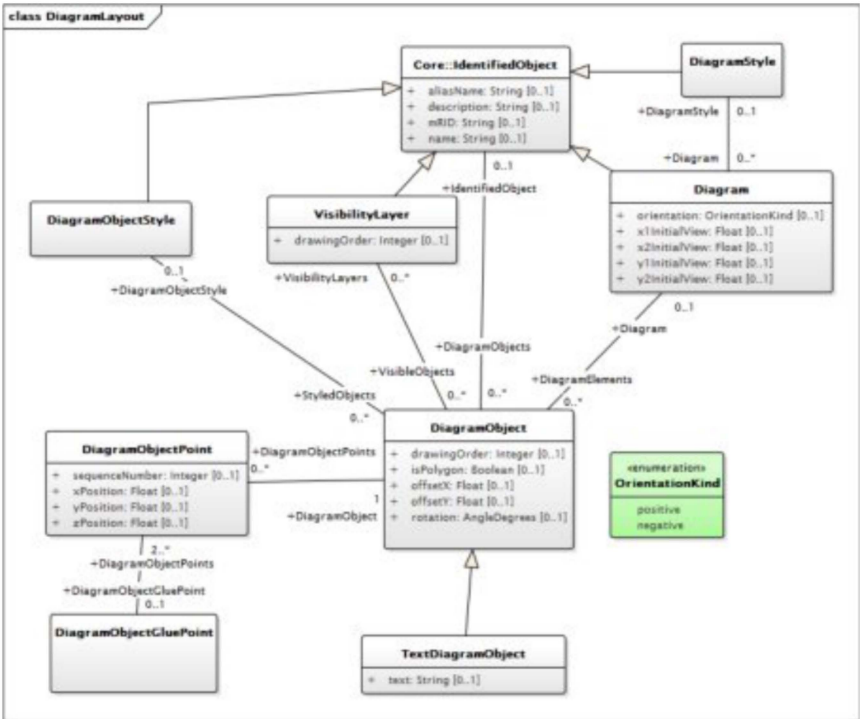
- A generic method for linking the diagram object to the domain data. Domain data and diagram layout data can be exchanged together with the domain data, or separately from each other with the assumption that domain data will have already been imported should diagram layout data be imported separately.
- It does not require or imply a specific domain data format. Therefore, it supports domain data modelled according to the IEC 61970-301 Common information model (CIM) that is exchanged in the IEC 61970-501 format (CIM RDF schema). Diagram layout data references domain data in compliance with IEC/TR 62541-1, *OPC Unified Architecture*.
- The intended usage of this standard is that the source of a diagram will use the standard to encode the layout of their diagram as they created it. It is then incumbent upon any receiver to supply the means of using that diagram layout in their system.
 - In the simplest situation, if the source uses Switch object placements, the receiver creates a generic Switch template in the receiver's system that will be used to render Switches (i.e. there is no exchange of Switch templates).
 - In more complex situations, a 1:1 correspondence may not be desirable, and receivers may have to create generic transformations in order to use exchanges.
 - However, in both situations, it is expected that once an investment is made in a strategy for rendering, repeated imports may be carried out automatically.
- Supports the exchange of diagram objects that have no relationship to domain data, i.e., pure static background objects.

- Supports multiple representations of the same domain object in the same or different diagrams.
- Allows the diagram to be used as the unit of exchange, providing a straightforward approach to partial exchange, or an exchange between systems that have a separate model and repository for diagram layout data.
- Supports assignment of diagram objects to layers or other means, for showing or hiding information based on zoom level and/or user interest.
- A generic method for proprietary extensions to enable full round tripping (export and import back into the same system) without information loss within a system, and without breaking the standard exchange format.

5.2 Diagram layout profile classes

5.2.1 General

The diagram layout classes are a part of IEC 61970-301 and as such are associated to the domain data using the standard UML association relationship. A new sub-package in IEC 61970-301 contains the classes required for expressing schematic data for domain objects as illustrated in Figure 5.



IEC

Figure 5 – Diagram layout information model

5.2.2 Diagram

5.2.2.1 General

The Diagram represents the diagram being exchanged. The initialView attributes can be used to specify an initial view with the x,y coordinates of the diagonal points. The coordinate system is a standard Cartesian coordinate system and the orientation attribute defines a positive or negative orientation. A positive orientation gives standard "right-hand" orientation, with negative orientation indicating a "left-hand"; orientation. For 2D diagrams, a positive orientation will result in X values increasing from left to right and Y values increasing from bottom to top. A negative orientation gives the "left-hand"; orientation (favoured by computer graphics displays) with X values increasing from left to right and Y values increasing from top to bottom.

5.2.2.2 Native members

orientation	0..1	OrientationKind	The Diagram represents the diagram being exchanged. The initialView attributes can be used to specify an initial view with the x,y coordinates of the diagonal points. The coordinate system is a standard Cartesian coordinate system and the orientation attribute defines a positive or negative orientation.
x1InitialView	0..1	Float	X coordinate of the first corner of the initial view
x2InitialView	0..1	Float	X coordinate of the second corner of the initial view
y1InitialView	0..1	Float	Y coordinate of the first corner of the initial view
y2InitialView	0..1	Float	Y coordinate of the second corner of the initial view

5.2.2.3 Inherited members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
aliasName	0..1	String	see IdentifiedObject
description	0..1	String	see IdentifiedObject

5.2.3 DiagramObject

5.2.3.1 General

The DiagramObject class defines an object that defines one or more points in a given space. This object can be associated with anything that subclasses *IdentifiedObject* in IEC 61970-301.

5.2.3.2 Native members

drawingOrder	0..1	Integer	The drawing order of this element. The higher the number, the later the element is drawn in sequence. This is used to ensure that elements that overlap are rendered in the correct order.
isPolygon	0..1	Boolean	Defines whether or not the diagram objects points define the boundaries of a polygon or the routing of a polyline. If this value is true then a receiving application should consider the first and last points to be connected.
offsetX	0..1	Float	The offset in the X direction. This is used for defining the offset from centre for rendering an icon (the default is that a single point specifies the centre of the icon). The offset is in per-unit with 0 indicating there is no offset from the horizontal centre of the icon. -0,5 indicates it is offset by 50 % to the left and 0,5 indicates an offset of 50 % to the right.
offsetY	0..1	Float	The offset in the Y direction. This is used for defining the offset from centre for rendering an icon (the default is that a single point specifies the centre of the icon). The offset is in per-unit with 0 indicating there is no offset from the vertical centre of the icon. The offset direction is dependent on the orientation of the diagram, with -0,5 and 0,5 indicating an offset of $\pm 50\%$ on the vertical axis.
rotation	0..1	AngleDegrees	Sets the angle of rotation (in Degrees) of the diagram object in a clockwise direction from the normal
Diagram	1..1	Diagram	A diagram object is part of a Diagram
DiagramObjectPoints	0..unbounded	DiagramObjectPoint	A diagram object can have 0 or more points to reflect its layout position, routing (for polylines) or boundary (for polygons)
DiagramObjectStyle	0..1	DiagramObjectStyle	A diagram object has a style associated that provides a reference for the style used in the originating system
IdentifiedObject	0..1	IdentifiedObject	The domain object that this diagram object is associated with
DiagramStyle	0..1	DiagramStyle	A Diagram may have a DiagramStyle.

5.2.3.3 Inherited members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
aliasName	0..1	String	see IdentifiedObject
description	0..1	String	see IdentifiedObject

5.2.4 DiagramObjectGluePoint

This is used for grouping DiagramObjectPoints from different DiagramObjects that are considered to be “glued” together in a diagram even if they are not at the exact same coordinates.

5.2.5 DiagramObjectPoint

5.2.5.1 General

A DiagramObjectPoint is a point in a given space defined by 3 coordinates and associated to a DiagramObject. The coordinates may be positive or negative as the origin does not have to be in the corner of a diagram. The sequence attribute is used when a DiagramObject has more than one DiagramObjectPoint in which case this defines the drawing order. A DiagramObject may represent any CIM object. For single line diagrams such objects can include:

- Analogue measurement values
- Breaker/disconnector
- Power transformer
- Transmission line

5.2.5.2 Native members

sequenceNumber	0..1	Integer	The sequence position of the point, used for defining the order of points for DiagramObjects acting as a polyline or polygon with more than one point
xPosition	1..1	Float	The X coordinate of this point
yPosition	1..1	Float	The Y coordinate of this point
zPosition	0..1	Float	The Z coordinate of this point
DiagramObject	1..1	DiagramObject	The diagram object with which the points are associated
DiagramObjectGluePoint	0..1	DiagramObjectGluePoint	The 'glue' point that this point is associated with

5.2.6 DiagramObjectStyle

5.2.6.1 General

DiagramObjectStyle is a reference to a style used by the originating system for a DiagramObject. A DiagramObjectStyle describes information such as:

- line thickness
- shape, e.g circle, rectangle ...
- colour

5.2.6.2 Inherited members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
aliasName	0..1	String	see IdentifiedObject
description	0..1	String	see IdentifiedObject

5.2.7 TextDiagramObject

5.2.7.1 General

A TextDiagramObject is a diagram object for placing free-text or text derived from an associated domain object.

5.2.7.2 Native members

text	0..1	String	The text that is displayed by this text diagram object
------	------	--------	--

5.2.7.3 Inherited members

drawingOrder	0..1	Integer	see DiagramObject
isPolygon	0..1	Boolean	see DiagramObject
offsetX	0..1	Float	see DiagramObject
offsetY	0..1	Float	see DiagramObject
rotation	0..1	AngleDegrees	see DiagramObject
Diagram	1..1	Diagram	see DiagramObject
DiagramObjectPoints	0..unbounded	DiagramObjectPoint	see DiagramObject
DiagramObjectStyle	0..1	DiagramObjectStyle	see DiagramObject
IdentifiedObject	0..1	IdentifiedObject	see DiagramObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
aliasName	0..1	String	see IdentifiedObject
description	0..1	String	see IdentifiedObject

5.2.8 VisibilityLayer**5.2.8.1 General**

Layers are typically used for grouping diagram objects according to themes and scales. Themes are used to display or hide certain information (e.g., lakes, borders), while scales are used for hiding or displaying information depending on the current zoom level (hide text when it is too small to be read, or when it exceeds the screen size). This is also called de-cluttering.

The diagram layout profile will support an m:n relationship between diagram objects and layers. It will be the task of the importing system to convert an m:n case into an appropriate 1:n representation if the importing system does not support m:n.

5.2.8.2 Native members

drawingOrder	0..1	Integer	The drawing order for this layer. As with the drawingOrder for diagram objects, the higher the number, the later the layer and the objects within it are rendered.
VisibleObjects	1..unbounded	DiagramObject	A visibility layer can contain one or more diagram objects

5.2.8.3 Inherited members

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
aliasName	0..1	String	see IdentifiedObject
description	0..1	String	see IdentifiedObject

5.2.9 Abstract classes – IdentifiedObject**5.2.9.1 General**

An IdentifiedObject is a root class to provide common identification for all classes needing identification and naming attributes.

5.2.9.2 Native members

name	mult	type	description
aliasName	0..1	String	The aliasName is free text human readable name of the object alternative to IdentifiedObject.name. It may be non unique and may not correlate to a naming hierarchy. The attribute aliasName is retained because of backwards compatibility between CIM releases. It is however recommended to replace aliasName with the Name class as aliasName is planned for retirement at a future time.
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.

5.2.10 Enumerations – OrientationKind

negative
positive

5.2.11 DiagramStyle**5.2.11.1 General**

The diagram style refers to a style used by the originating system for a diagram. A diagram style describes information such as schematic, geographic, bus-branch etc.

5.2.11.2 Native members

Diagram	0..*	Diagram	A DiagramStyle can be used by many Diagrams.
---------	------	---------	--

5.2.11.3 Inherited members

aliasName	0..1	String	see IdentifiedObject
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

6 Graphical rendering

6.1 General

Section 6 illustrates the use of the object model described in section 5. It uses the conventions shown in Figure 6 for representing object instances and relationships.

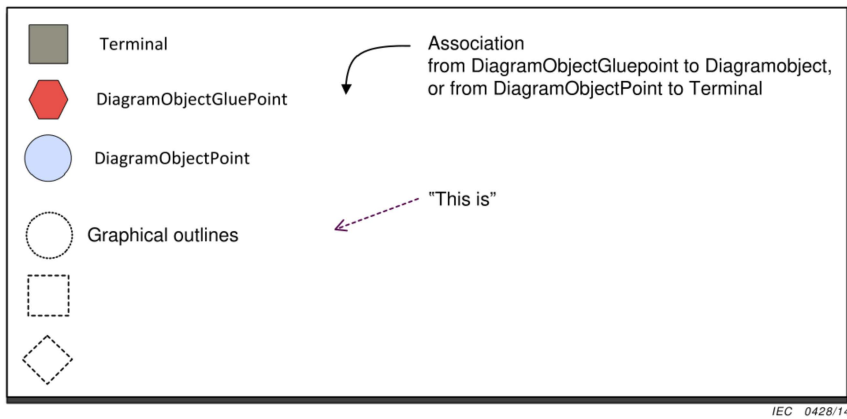
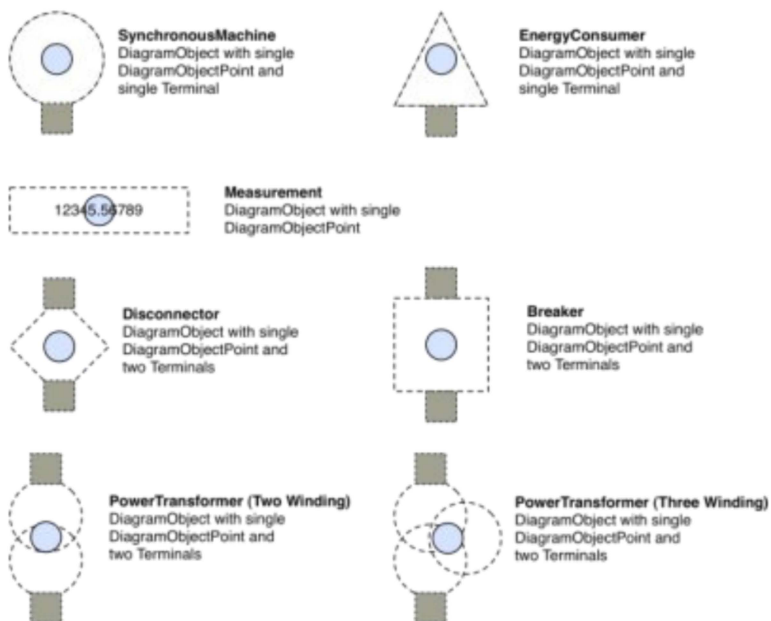


Figure 6 – Conventions used for representing object instances and associations

6.2 Single point objects

The simplest Diagram Objects have a single point in the form of a DiagramObjectPoint. These are used for specifying the centre-point of an icon, which can be pieces of network equipment or measurement locations on a diagram. This is shown in Figure 7.

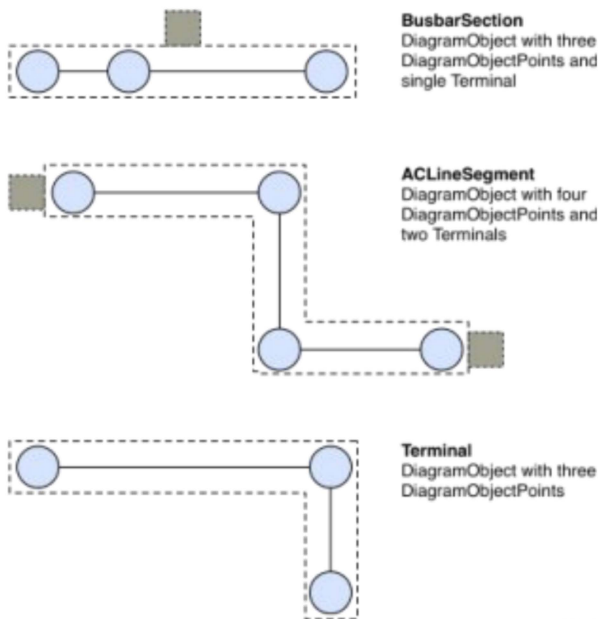


IEC 0429/14

Figure 7 – Single point diagram objects

6.3 Multiple point objects

For those diagram objects that cannot be accurately rendered using a single point, multiple DiagramObjectPoints are used within a single DiagramObject. Terminals themselves can be given multiple points so that the interconnection of equipment can be displayed correctly as shown in Figure 8.

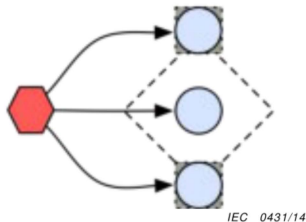


IEC 0430/14

Figure 8 – Multiple point diagram objects

6.4 Gluing points

DiagramObjectGluePoints are used to identify when two or more points on different DiagramObjects are considered to be 'glued' together on a diagram. This information is required so that a receiver can identify the case of multiple, synchronised points, which may not be at the same coordinates and thus ensure that modifications to one are reflected in the other points. An example shown in Figure 9 has a Disconnecter with two Terminals and three DiagramObjectPoints, all part of different DiagramObjects since each Terminal and the Disconnecter are separate objects in IEC 61970-301.



IEC 0431/14

Figure 9 – Disconnecter with glue point

The glue point here is used to identify that the three DiagramObjectPoints are related to each other even though they are at different coordinates.

In the electrical topology model defined in IEC 61970-301 there is a ConnectivityNode object at this point. However its inclusion here would be superfluous as the DiagramObjectGluePoint provides a mechanism for identifying that these DiagramObjectPoints are connected, as illustrated in Figure 10.

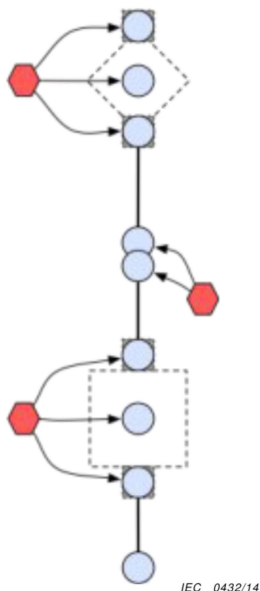


Figure 10 – Disconnector and breaker with glue points

6.5 Diagram object style

The diagram object style defines how to convert the state of a domain object into a visible representation, using information such as:

- line thickness
- shape, e.g circle, rectangle
- colour

The definition of the styles and their use for graphical rendering are typically solved in very specific ways for each system and are not part of this standard. The diagram layout transfers only make references to the diagram object style, not the definition of the diagram object style itself. The importing system will have to map the imported references to its best matching local diagram object style. This shall be by an agreement between the sending and receiving systems, which is outside of this standard's scope.

As an example, if the importing system can derive that a domain object is representing a remote controlled pole mounted load break switch with alarm indication, this might be enough to establish a sufficiently identical visualisation in the importing system.

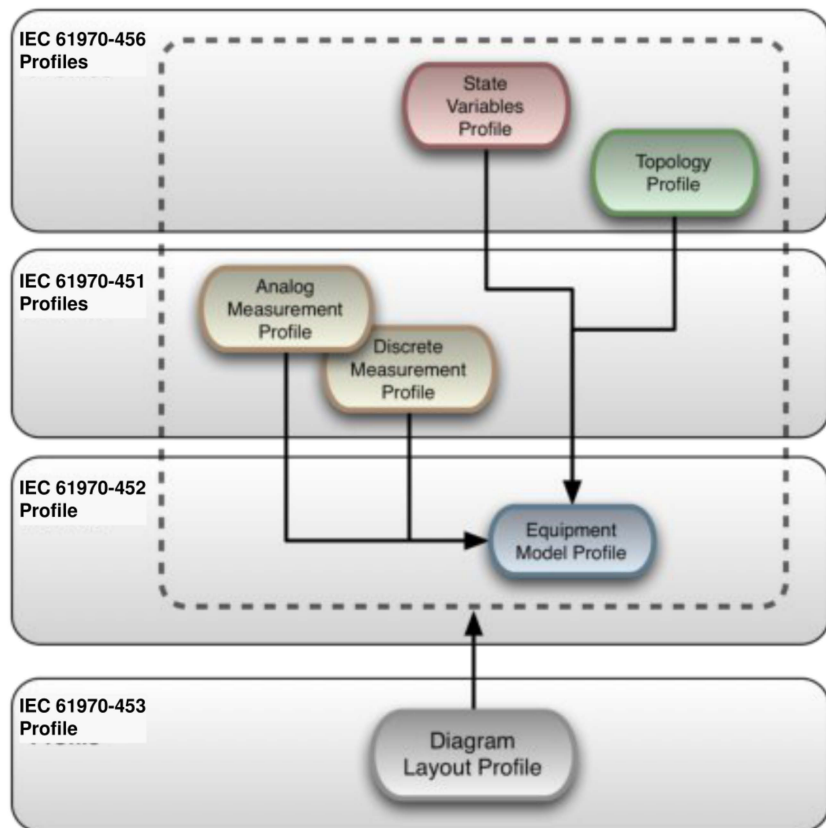
6.6 Diagram layout exchange process

Diagram layout exchanges allow the exchanging of diagram layout data independent of and separate from domain data. Using the approach of separating data into profiles for specific

types of data exchanges, the diagram layout profile detailed here can be utilised along with other profiles defined in other IEC standards.

IEC 61970-452 defines profiles for the exchange of power system models with an equipment profile defining electrical equipment and connectivity. IEC 61970-456 defines profiles for measurements, state variables and topology data that are dependent on the equipment profile defined in IEC 61970-452. This standard uses the same approach with the diagram layout profile utilising the same Equipment profile.

Figure 11 gives an overview of these profiles:



IEC 0433/14

Figure 11 – Profiles within IEC standards

When importing diagram layout data separate from domain (i.e. Equipment) data, the following rules apply.

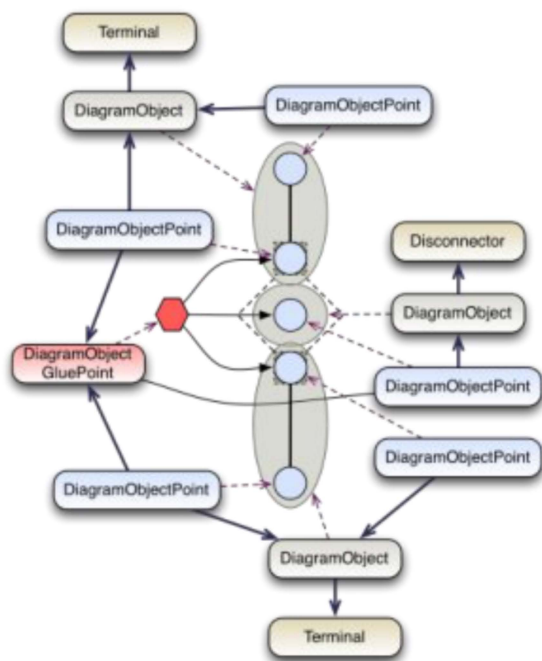
- The domain data import has to take place before the diagram layout import, otherwise it is not possible to resolve references from diagram objects to domain objects.
- It is the task of the importing program to reasonably report and handle inconsistencies between diagram objects and domain data, for example missing domain data.

For instance data exchange using CIM XML encoding as defined in IEC 61970-552, the header information will define the unique identifier of the data set on which the schematic layout data depends, thus allowing an importing system to determine whether it can successfully process the data.

7 Examples

7.1 Data instantiation and encoding

Taking a single disconnector with two terminals the diagrammatical view consists of eleven objects using five classes from the UML: the *Disconnector*, two *Terminals*, three *DiagramObjects*, five *DiagramObjectPoints* and a single *DiagramObjectGluePoint*. This is illustrated in Figure 12. This diagram shows the objects created, the associations between them and their corresponding position in the diagrammatical view used in the previous figures.



IEC 0434/14

Figure 12 – Disconnector diagram object instantiation

When encoded in the format defined by IEC 61970-552 the resulting XML is shown in Figure 13:

```
<cim:DiagramObjectGluePoint rdf:ID="gp0">
</cim:DiagramObjectGluePoint>

<cim:Disconnecter rdf:ID="discon0">
  <cim:IdentifiedObject.name>Disconnecter</cim:IdentifiedObject.name>
</cim:Disconnecter>
<cim:DiagramObject rdf:ID="disconDO">
  <cim:DiagramObject.IdentifiedObject rdf:resource="#discon0"/>
</cim:DiagramObject>
<cim:DiagramObjectPoint rdf:ID="disconDOP0">
  <cim:DiagramObjectPoint.xPosition>10</cim:DiagramObjectPoint.xPosition>
  <cim:DiagramObjectPoint.yPosition>10</cim:DiagramObjectPoint.yPosition>
  <cim:DiagramObjectPoint.DiagramObject rdf:resource="#disconDO"/>
  <cim:DiagramObject.GluePoint rdf:resource="#gp0"/>
</cim:DiagramObjectPoint>

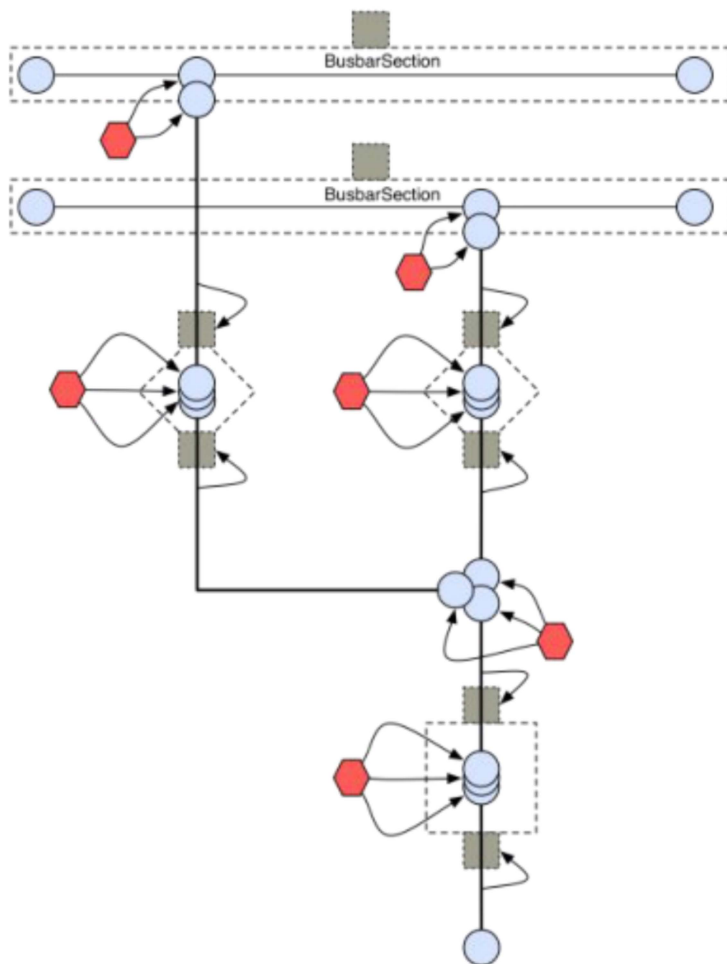
<cim:Terminal rdf:ID="t0">
  <cim:IdentifiedObject.name>Terminal 0</cim:IdentifiedObject.name>
  <cim:Terminal.ConductingEquipment rdf:resource="#discon0"/>
</cim:Terminal>
<cim:DiagramObject rdf:ID="t0DO">
  <cim:DiagramObject.IdentifiedObject rdf:resource="#t0"/>
</cim:DiagramObject>
<cim:DiagramObjectPoint rdf:ID="t0DOP0">
  <cim:DiagramObjectPoint.xPosition>10</cim:DiagramObjectPoint.xPosition>
  <cim:DiagramObjectPoint.yPosition>8</cim:DiagramObjectPoint.yPosition>
  <cim:DiagramObjectPoint.DiagramObject rdf:resource="#t0DO"/>
  <cim:DiagramObjectPoint.DiagramObjectGluePoint rdf:resource="#gp0"/>
</cim:DiagramObjectPoint>
<cim:DiagramObjectPoint rdf:ID="t0DOP1">
  <cim:DiagramObjectPoint.xPosition>10</cim:DiagramObjectPoint.xPosition>
  <cim:DiagramObjectPoint.yPosition>6</cim:DiagramObjectPoint.yPosition>
  <cim:DiagramObjectPoint.DiagramObject rdf:resource="#t0DO"/>
</cim:DiagramObjectPoint>

<cim:Terminal rdf:ID="t1">
  <cim:IdentifiedObject.name>Terminal 0</cim:IdentifiedObject.name>
  <cim:Terminal.ConductingEquipment rdf:resource="#discon0"/>
</cim:Terminal>
<cim:DiagramObject rdf:ID="t1DO">
  <cim:DiagramObject.IdentifiedObject rdf:resource="#t1"/>
</cim:DiagramObject>
<cim:DiagramObjectPoint rdf:ID="t1DOP0">
  <cim:DiagramObjectPoint.xPosition>10</cim:DiagramObjectPoint.xPosition>
  <cim:DiagramObjectPoint.yPosition>12</cim:DiagramObjectPoint.yPosition>
  <cim:DiagramObjectPoint.DiagramObject rdf:resource="#t1DO"/>
  <cim:DiagramObjectPoint.DiagramObjectGluePoint rdf:resource="#gp0"/>
</cim:DiagramObjectPoint>
<cim:DiagramObjectPoint rdf:ID="t1DOP1">
  <cim:DiagramObjectPoint.xPosition>10</cim:DiagramObjectPoint.xPosition>
  <cim:DiagramObjectPoint.yPosition>16</cim:DiagramObjectPoint.yPosition>
  <cim:DiagramObjectPoint.DiagramObject rdf:resource="#t1DO"/>
</cim:DiagramObjectPoint>
```

Figure 13 – IEC 61970-552 Encoding for disconnecter diagram data

7.2 Simple bay example use case

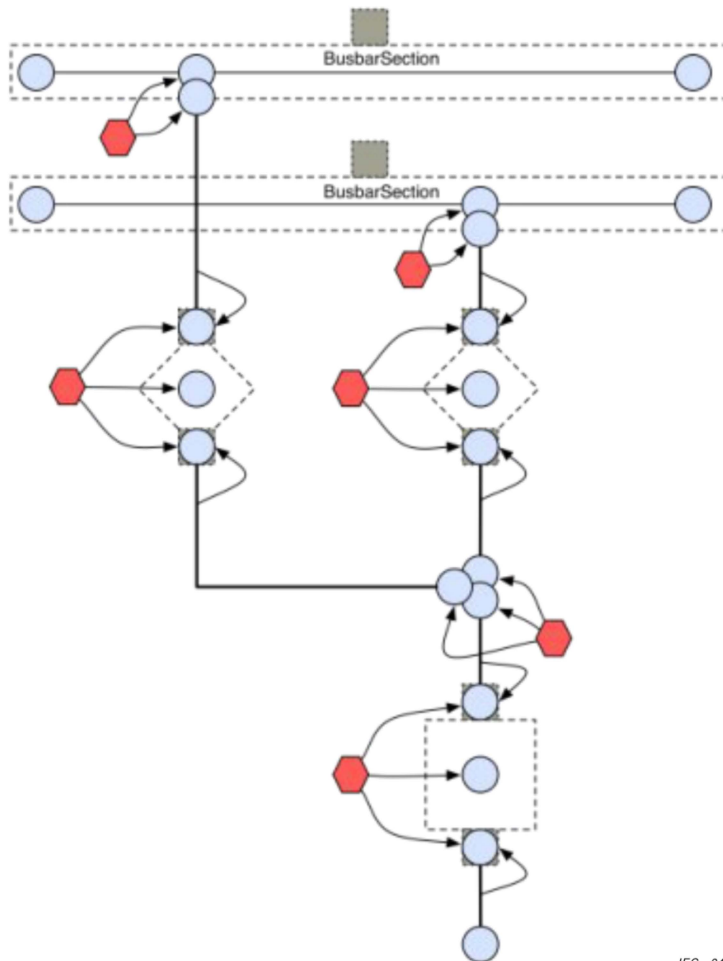
Figure 14 illustrates how to apply the exchange model to the GIS construction type simple bay example use case (lines drawn to the centre point).



IEC 0435/14

Figure 14 – Bay diagram example with objects outlined GIS style

Figure 15 illustrates how to apply the exchange model to the SCADA construction type simple bay example use case (lines drawn to the terminals).



IEC 0436/14

Figure 15 – Bay diagram example with objects outlined SCADA/EMS style

Annex A (informative)

Benefits and format conversion from IEC 61970-453 Edition 1 to Edition 2

This edition includes the following significant technical changes with respect to the previous edition:

- a) The SVG elements and its data model have been replaced by the Diagram Layout Package, which is now an integral part of the IEC 61970-301 (CIM) model.
- b) The exchange is in accordance with and is a part of the IEC 61970 profile concept.
- c) A glue point object has been introduced to model explicit connections between graphics elements.

Benefits resulting from the changes made to Edition 2:

- Clear separation between layout and rendering eliminates ambiguities in the use of SVG and style features in Edition 1.
- The same encoding can be used as for other IEC 61970-301 packages, enabling the exchange either as a separate document, or combined with data from other IEC 61970-301 packages.
- The glue point allows the resolution of differences in rendering between the sending and the receiving system, and it supports editing in the receiving system without losing connections between graphic elements.

The basic concept of separating layout, rendering and domain data has not been changed. It should be possible to convert the format from Edition 1 to Edition 2 to a large extent by using transformation rules. A summary of the changes is shown in Table A.1.

Table A.1 – Summary of changes between Edition 1 and Edition 2

Feature	Edition 1	Edition 2
Schematic diagram	CgoDiagram (SVG element)	Diagram
Graphical Object within a diagram	CgoGraphicalObject, CgoSymbol, CgoBasicShape (SVG elements)	DiagramObject, TextDiagramObject, DiagramObjectPoint
Element grouping	CgoGroup	<not supported>
References	Metadata (SVG concept)	Explicit relationship model
Rendering reference	Metadata PresentationLogic_Ref	DiagramObjectStyle
Layer membership	Metadata Layer_Ref	VisibilityLayer
Domain data reference	Metadata PSR_Ref, Meas_Ref	IdentifiedObject
User Interaction	Metadata ComplexObject	<not supported>
Predefined views	CgoDiagramView	<not supported>
Graphic connection	<not supported>	DiagramObjectGluePoint

Bibliography

IEC 61968-11, *Application integration at electric utilities – System interfaces for distribution management – Part 11: Common information model (CIM) extensions for distribution*

IEC 61970-1, *Energy management system application program interface (EMS-API) – Part 1: Guidelines and general requirements*

IEC/TS 61970-2, *Energy management system application program interface (EMS-API) – Part 2: Glossary*

IEC 61970-452, *Energy management system application program interface (EMS-API) – Part 452: CIM Static transmission network model profiles*

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch